

CCNA-LEVEL • FULL COURSE GUIDE

Switching & Routing Mastery

A complete, exam-oriented guide to Cisco switching and routing — concepts, configurations, diagrams, hands-on labs, and practice questions for CCNA 200-301 preparation.

27 Chapters

40+ Labs & Configs

Diagrams & Tables

Practice Questions

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Networking Foundations Refresher



LEARN WITH ABHIRUP

1 The OSI & TCP/IP Models

Before diving into switching and routing, you need a rock-solid mental model of how data actually moves across a network. The OSI (Open Systems Interconnection) model breaks this journey into seven layers. It's rarely implemented exactly as written, but it is the shared vocabulary every networking exam, vendor, and engineer uses to describe where a problem lives.

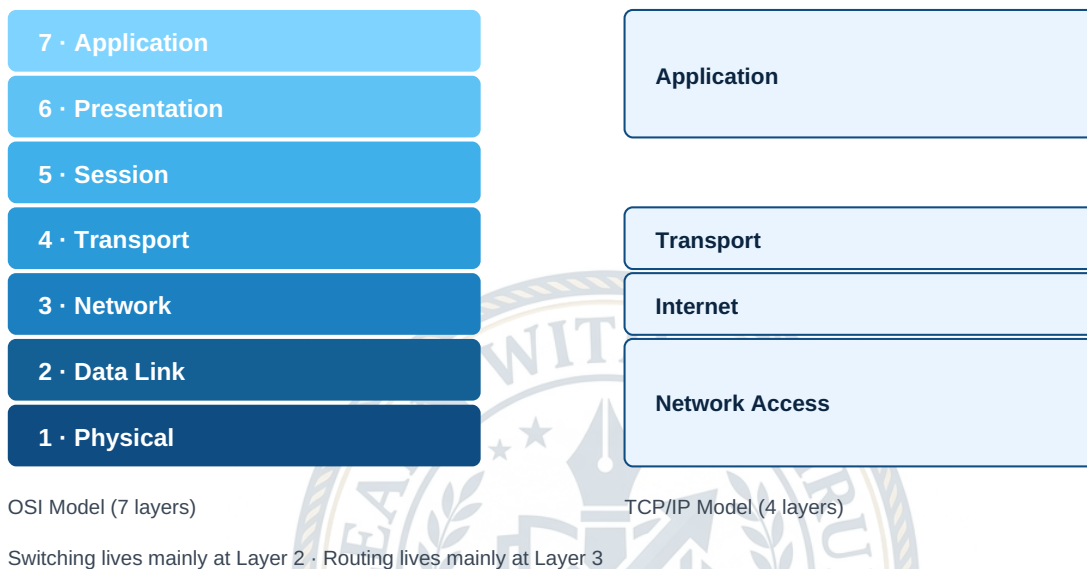


Figure 1.1 — The OSI seven-layer model mapped against the practical TCP/IP four-layer model.

Why this matters for switches and routers

Switches are fundamentally **Layer 2** devices — they forward frames based on MAC addresses. Routers are fundamentally **Layer 3** devices — they forward packets based on IP addresses. Modern "Layer 3 switches" blur this line by doing both jobs in the same box, but the exam and real troubleshooting always start with the question: "Is this a Layer 2 problem or a Layer 3 problem?"

Layer	Name	PDU	Key Devices / Protocols
7	Application	Data	HTTP, DNS, DHCP, SSH
6	Presentation	Data	TLS/SSL, encoding, encryption
5	Session	Data	Session establishment (e.g., NetBIOS)
4	Transport	Segment	TCP, UDP
3	Network	Packet	IP, ICMP, Routers
2	Data Link	Frame	Ethernet, MAC, Switches
1	Physical	Bits	Cables, hubs, NICs, connectors

EXAM TIP

The CCNA loves to test PDU names. Memorize: Data → Segment → Packet → Frame → Bits (top to bottom). When a question says "encapsulation," it's moving down the stack; "de-encapsulation" moves up.

Encapsulation walk-through

When a PC sends an HTTP request, the data is wrapped in layers exactly like a package inside boxes:

1. The web browser hands data to the OS as an HTTP request (Application layer).
2. TCP adds a header with source/destination **port numbers** (e.g., 443) — now it's a **segment**.
3. IP adds a header with source/destination **IP addresses** — now it's a **packet**.
4. Ethernet adds a header/trailer with source/destination **MAC addresses** — now it's a **frame**.
5. The NIC converts the frame into electrical signals or light pulses — **bits** on the wire.

NOTE

A switch never looks at IP addresses to make a forwarding decision (unless it's a multilayer/Layer-3 switch acting as a router). A router never looks at MAC addresses to decide *where* to send a packet across the network — it only uses MAC addresses for the very next hop.



2 IPv4 Addressing & Subnetting Refresher

Every switching and routing decision downstream depends on being fast and accurate with IPv4 addressing. This chapter is a dense refresher — if subnetting already feels natural to you, skim the tables; if it doesn't, work through the examples with a pen.

Address classes and private ranges

Class	First Octet Range	Default Mask	Usable Hosts
A	1 – 126	/8 (255.0.0.0)	16,777,214
B	128 – 191	/16 (255.255.0.0)	65,534
C	192 – 223	/24 (255.255.255.0)	254

PRIVATE IP RANGES (RFC 1918)

10.0.0.0/8 • 172.16.0.0/12 • 192.168.0.0/16 — these are never routed on the public internet and require NAT to reach it (covered in Chapter 20).

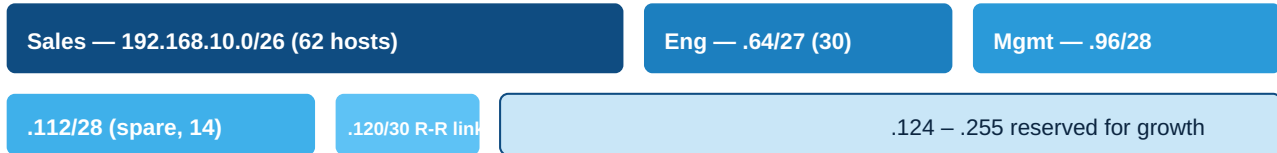
The subnet mask cheat sheet

Memorize this table cold. It is the single highest-leverage piece of knowledge for the exam and for real VLSM design.

CIDR	Mask	Hosts (/subnet)	Block Size
/24	255.255.255.0	254	256
/25	255.255.255.128	126	128
/26	255.255.255.192	62	64
/27	255.255.255.224	30	32
/28	255.255.255.240	14	16
/29	255.255.255.248	6	8
/30	255.255.255.252	2 (point-to-point link)	4

Worked example — VLSM design

Suppose you're given `192.168.10.0/24` and need to design subnets for: Sales (60 hosts), Engineering (28 hosts), Management (12 hosts), and a router-to-router link (2 hosts). Variable Length Subnet Masking lets you size each subnet to fit, instead of wasting addresses with one fixed mask everywhere.



Allocation logic: always assign the largest subnet first, then fill remaining space.

Sales needs 60 → next power-of-2 block is 64 → /26 (`192.168.10.0 – .63`)

Engineering needs 28 → next block is 32 → /27 (`192.168.10.64 – .95`)

Management needs 12 → next block is 16 → /28 (`192.168.10.96 – .111`)

Figure 2.1 — VLSM allocation carving one /24 into right-sized subnets, largest to smallest.

EXAM TIP — THE SUBNETTING FORMULA

Usable hosts = $2^h - 2$, where h = number of host bits. Number of subnets = 2^s , where s = number of borrowed bits. The -2 accounts for the network address (all host bits 0) and broadcast address (all host bits 1) — neither is assignable to a device.

Quick subnetting drill

Given `172.16.50.77 /28`, find the network address, broadcast address, and valid host range.

```
Block size for /28 = 256 - 240 = 16
Multiples of 16 near 77: 64, 80
Network address    = 172.16.50.64
Broadcast address  = 172.16.50.79
Valid host range   = 172.16.50.65 - 172.16.50.78
```

PRACTICE DRILL

Work these out on paper before checking Chapter 27's answer key:

- 1) `10.1.1.200/26` → network, broadcast, host range?
- 2) `192.168.5.130/27` → network, broadcast, host range?
- 3) How many usable hosts in a /29? How many /29 subnets fit in a /24?

3 Network Devices & Cabling

A quick device refresher so terminology in later chapters is unambiguous.

Device	Layer	Forwarding Decision Based On
Hub (legacy)	1	None — floods everything, one collision domain
Switch	2	Destination MAC address
Router	3	Destination IP address / routing table
Layer 3 Switch	2 & 3	Both — switches within a VLAN, routes between VLANs
Access Point	1/2	Bridges wireless clients into the wired LAN

Cabling standards

- **Straight-through cable** — connects unlike devices (PC → switch, switch → router). Modern switches auto-detect with Auto-MDIX, but know the rule for the exam.
- **Crossover cable** — connects like devices (switch → switch, PC → PC, router → router) on older hardware without Auto-MDIX.
- **Fiber (single-mode/multi-mode)** — used for longer runs and higher bandwidth backbone/uplink connections between switches and between buildings.

NOTE

Collision domains vs. broadcast domains: every switch port is its own collision domain (a huge improvement over hubs). A broadcast domain is bounded by a router interface (or a VLAN boundary) — this distinction becomes crucial in Chapter 6 onward.

Switching Fundamentals



4 How Switches Work

A switch's entire job comes down to one loop: **learn, forward, flood, age**. Understanding this loop deeply makes VLANs, trunking, and STP click into place naturally later.

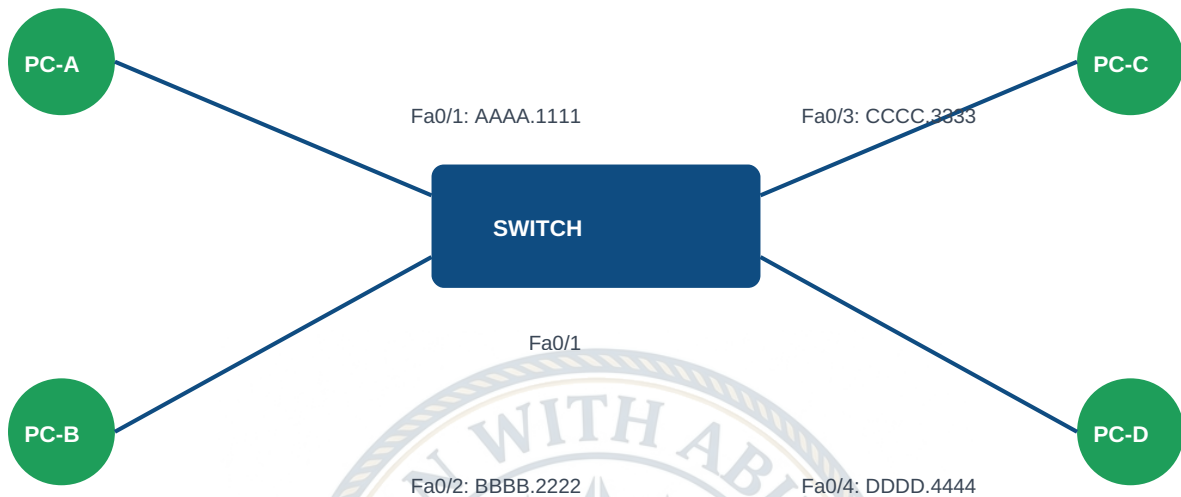


Figure 4.1 — Four hosts on four switch ports; the switch builds a MAC address table mapping each MAC to its port.

The learn / forward / flood / age process

1. **Learn:** When a frame enters a port, the switch records the *source* MAC address and the port it arrived on in the MAC address table (also called the CAM table).
2. **Forward or Flood:** The switch looks up the *destination* MAC. If it's in the table, the frame is forwarded out only that port. If it's unknown, the frame is flooded out every port except the one it arrived on.
3. **Age:** Entries that see no traffic for the aging timer (default 300 seconds) are removed, keeping the table current as devices move or disconnect.

```
! Viewing the MAC address table
Switch# show mac address-table
      Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	aaaa.1111.aaaa	DYNAMIC	Fa0/1
1	bbbb.2222.bbbb	DYNAMIC	Fa0/2
1	cccc.3333.cccc	DYNAMIC	Fa0/3
1	dddd.4444.dddd	DYNAMIC	Fa0/4

EXAM TIP

Three special frame types are always flooded regardless of the MAC table: **broadcasts** (FFFF.FFFF.FFFF), **multicasts**, and **unknown unicasts** (destination not yet learned). Together these are sometimes abbreviated "BUM" traffic.

Switching methods

Method	Behavior	Latency	Error Checking
Store-and-forward	Buffers entire frame before forwarding	Higher	Full CRC check — drops bad frames
Cut-through	Forwards after reading destination MAC only	Lowest	None — can forward corrupt frames
Fragment-free	Reads first 64 bytes (past collision window) before forwarding	Medium	Catches most collision fragments



5 Switch Port Configuration Basics

Before touching VLANs or STP, you need fluency in basic interface configuration — this is the muscle memory the exam (and real job) demands.

```
! Enter interface configuration mode and set speed/duplex
Switch(config)# interface fastethernet 0/1
Switch(config-if)# speed 100
Switch(config-if)# duplex full
Switch(config-if)# description Link-to-PC-A
Switch(config-if)# no shutdown
```

COMMON PITFALL

A **duplex mismatch** (one side full, one side half) is one of the most common real-world causes of intermittent slowness and late collisions. Always match speed and duplex on both ends, or set both to `auto`.

Naming conventions across platforms

Interface Type	Example Name	Typical Speed
Ethernet	Ethernet0/1	10 Mbps (legacy)
FastEthernet	FastEthernet0/1 (Fa0/1)	100 Mbps
GigabitEthernet	GigabitEthernet0/1 (Gi0/1)	1000 Mbps
TenGigabitEthernet	TenGigabitEthernet0/1 (Te0/1)	10 Gbps

Useful verification commands

```
Switch# show interfaces Fa0/1 status
Switch# show interfaces Fa0/1
Switch# show running-config interface Fa0/1
Switch# show ip interface brief
```

LAB 5.1 — BASIC SWITCH BRING-UP

1. Configure a hostname and enable secret on SW1.
2. Set the management VLAN 1 SVI IP to 192.168.1.2/24 and bring it up.
3. Set a default gateway with `ip default-gateway 192.168.1.1`.
4. Configure Fa0/1–0/4 with descriptive labels and verify with `show ip interface brief`.

6 VLANs — Virtual LANs

A VLAN is a logical broadcast domain that can span one switch or many, independent of physical location. Instead of one giant flat network where every broadcast reaches every device, VLANs let you segment traffic by department, function, or security zone — all on the same physical switch hardware.

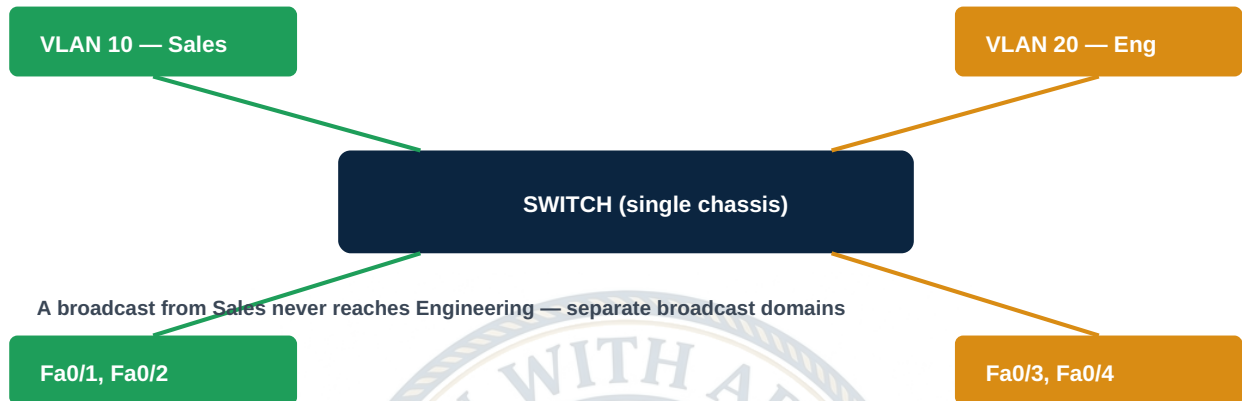


Figure 6.1 — Two VLANs on a single physical switch, each its own broadcast domain.

Why segment with VLANs?

- **Security** — isolate sensitive traffic (finance, servers) from general user traffic.
- **Broadcast control** — smaller broadcast domains mean less flooding and better performance.
- **Flexibility** — group users logically (by department) rather than by physical wiring closet.
- **Simplified management** — apply policy per VLAN instead of per physical port.

VLAN ranges

Range	Type	Notes
0, 4095	Reserved	Not usable
1	Default	Cannot be deleted or renamed; carries CDP/VTP/STP by default
2 – 1001	Normal range	Stored in vlan.dat; usable on all switches
1002 – 1005	Reserved	Legacy FDDI/Token Ring
1006 – 4094	Extended range	Requires VTP transparent mode on older platforms

Configuring VLANs

```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Engineering
Switch(config-vlan)# exit

! Assign access ports to a VLAN
Switch(config)# interface range Fa0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

Verification

```
Switch# show vlan brief
VLAN Name      Status      Ports
-----
1    default      active      Fa0/5, Fa0/6, Fa0/7, Fa0/8
10   Sales        active      Fa0/1, Fa0/2
20   Engineering   active      Fa0/3, Fa0/4
```

EXAM TIP

Deleting a VLAN that still has ports assigned to it does not delete those ports — they become "inactive" (not forwarding) until reassigned to an existing VLAN. Always reassign ports before removing a VLAN.

LAB 6.1 — BASIC VLAN SEGMENTATION

1. Create VLAN 10 (Sales), VLAN 20 (Engineering), VLAN 99 (Management) on SW1.
2. Assign Fa0/1–0/4 to VLAN 10, Fa0/5–0/8 to VLAN 20.
3. Confirm two PCs in different VLANs cannot ping each other yet (expected — no routing configured).
4. Confirm two PCs in the *same* VLAN can ping each other.

7 Trunking (802.1Q) & VTP

A single access link can only carry one VLAN. To carry traffic for *multiple* VLANs across one physical link — typically between switches — you need a **trunk**. Trunks tag each frame with its VLAN ID so the receiving switch knows which VLAN it belongs to.

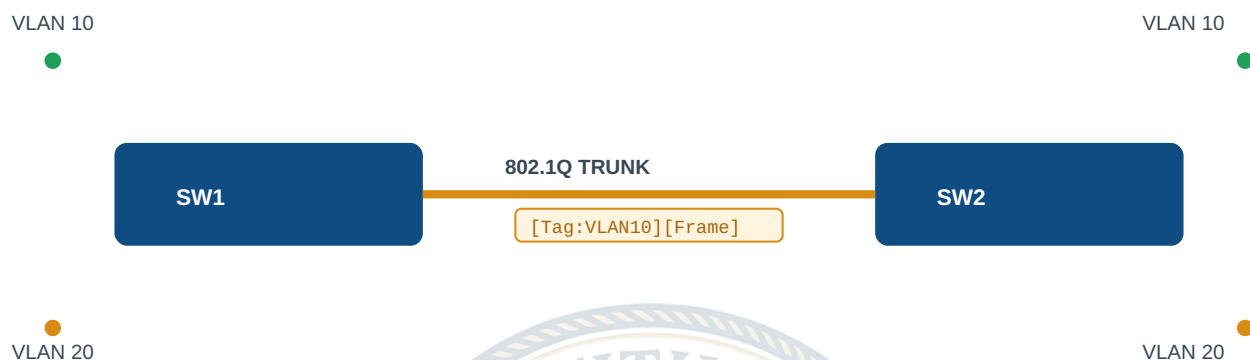


Figure 7.1 — One 802.1Q trunk carrying tagged frames for VLAN 10 and VLAN 20 between two switches.

How 802.1Q tagging works

802.1Q inserts a 4-byte tag into the Ethernet frame header containing the VLAN ID (12 bits, supporting VLANs 1–4094) and a priority field (for QoS). The **native VLAN** (default VLAN 1) is the one exception — frames in the native VLAN are sent *untagged* across the trunk for backward compatibility.

NATIVE VLAN MISMATCH

If the native VLAN differs on each end of a trunk, untagged traffic gets misassigned to the wrong VLAN, and the switch logs a native VLAN mismatch error. Always match native VLANs on both trunk ends — and many security best-practices recommend changing the native VLAN away from VLAN 1 entirely.

Configuring a trunk

```
Switch(config)# interface gigabitethernet 0/1
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk native vlan 99
Switch(config-if)# switchport trunk allowed vlan 10,20,99
```



```
Switch# show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gi0/1     on        802.1q         trunking    99
Port      Vlans allowed on trunk
Gi0/1     10,20,99
```

Dynamic Trunking Protocol (DTP)

Mode	Behavior
<code>switchport mode access</code>	Never trunks
<code>switchport mode trunk</code>	Always trunks
<code>switchport mode dynamic auto</code>	Trunks only if the other side actively wants to
<code>switchport mode dynamic desirable</code>	Actively tries to negotiate a trunk

EXAM TIP

Best practice — and common exam answer — is to manually hard-code `switchport mode trunk` or `switchport mode access` everywhere and disable DTP negotiation with `switchport nonegotiate`. Leaving ports on "dynamic auto/desirable" is a known VLAN-hopping security risk.

VTP (VLAN Trunking Protocol)

VTP propagates VLAN database information across a domain of trunked switches so you don't have to manually create every VLAN on every switch. It has three modes:

Mode	Can Create/Edit VLANs?	Forwards VTP Ads?
Server	Yes	Yes
Client	No	Yes
Transparent	Yes (locally only)	Yes (passes through, doesn't apply)

REAL-WORLD WARNING

VTP's revision-number mechanism has caused major outages: a switch with an old configuration but a *higher* revision number can accidentally wipe VLAN databases domain-wide when connected. Many organizations now run every switch in **VTP transparent** mode, or use **VTPv3**, to avoid this entirely.

VLANs isolate broadcast domains — but eventually Sales needs to talk to Engineering. Since a switch alone cannot route between VLANs, you need Layer 3 involvement. There are three standard approaches.

Method 1 — Router-on-a-Stick

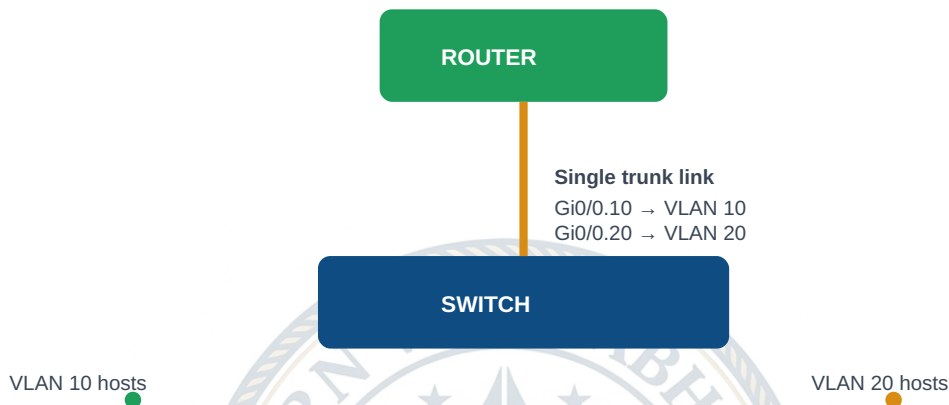


Figure 8.1 — Router-on-a-stick: one physical interface, divided into logical sub-interfaces, one per VLAN.

```
! Router configuration – sub-interfaces
Router(config)# interface gigabitethernet 0/0
Router(config-if)# no shutdown
Router(config)# interface gigabitethernet 0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gigabitethernet 0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0

! Switch side – the port facing the router must be a trunk
Switch(config)# interface gi0/1
Switch(config-if)# switchport mode trunk
```

BOTTLENECK ALERT

Router-on-a-stick funnels ALL inter-VLAN traffic through one physical link and the router's forwarding engine — fine for small branch offices, but a bandwidth chokepoint at scale. This is why enterprise LANs prefer method 2 below.

Method 2 — Switched Virtual Interfaces (SVI) on a Layer 3 Switch

A multilayer switch can route between VLANs internally at wire speed using SVIs — no external router, no trunk bottleneck.

```
L3Switch(config)# ip routing
L3Switch(config)# interface vlan 10
L3Switch(config-if)# ip address 192.168.10.1 255.255.255.0
L3Switch(config-if)# no shutdown
L3Switch(config)# interface vlan 20
L3Switch(config-if)# ip address 192.168.20.1 255.255.255.0
L3Switch(config-if)# no shutdown
```

NOTE

Don't forget **ip routing** — without it, a multilayer switch behaves purely as a Layer 2 device and will not route between SVIs even if they're configured.

Method 3 — Routed Ports

Any switch port can be converted into a pure Layer 3 routed port (losing its switching function) using **no switchport**. This is common for point-to-point links between two Layer 3 switches or to a router/WAN edge.

```
L3Switch(config)# interface gi0/1
L3Switch(config-if)# no switchport
L3Switch(config-if)# ip address 10.1.1.1 255.255.255.252
```

Method	Best For	Throughput
Router-on-a-stick	Small branch, few VLANs, budget constraints	Limited by trunk link
SVI on L3 switch	Enterprise LAN, campus core/distribution	Wire speed
Routed port	Point-to-point uplinks between L3 devices	Wire speed

LAB 8.1 — INTER-VLAN ROUTING WITH SVIS

1. Enable **ip routing** on a multilayer switch.
2. Create VLAN 10 and 20 SVIs with the addresses above.
3. Assign access ports to each VLAN.
4. Set each PC's default gateway to its SVI address, then verify full reachability with **ping** and **show ip route**.

Spanning Tree Protocol (STP/RSTP)

Redundant links between switches prevent single points of failure — but redundant Layer 2 links without control cause **broadcast storms**, MAC table instability, and duplicate frames, because Ethernet frames have no TTL to expire looping traffic. STP solves this by logically blocking redundant paths until they're needed.

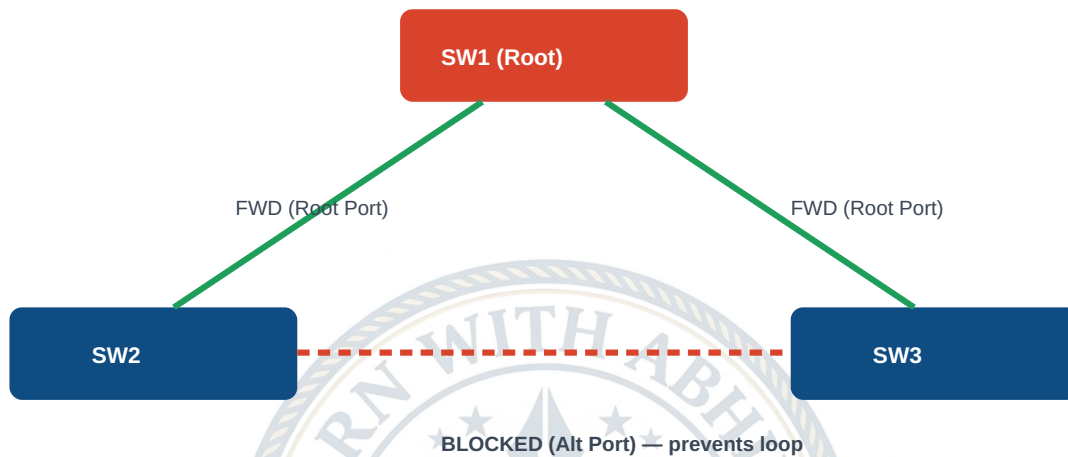


Figure 9.1 — SW1 is elected root bridge; the link between SW2 and SW3 is logically blocked to prevent a loop, while still available as backup.

Root bridge election

All switches exchange BPDUs (Bridge Protocol Data Units) to elect a single **root bridge** — the switch with the lowest Bridge ID (Priority + MAC address). Default priority is 32768. Every other switch then calculates its shortest cumulative path cost back to the root.

Port Role	Meaning
Root Port (RP)	One per non-root switch — best path back to the root bridge
Designated Port (DP)	One per segment — forwards traffic onto that segment
Blocking / Alternate Port	Not forwarding — prevents a loop, becomes active if the primary path fails

STP port states (802.1D classic)

State	Forwards Data?	Learns MACs?	Typical Duration
Blocking	No	No	Until topology change
Listening	No	No	15 sec
Learning	No	Yes	15 sec
Forwarding	Yes	Yes	Steady state
Disabled	No	No	Admin shutdown

CONVERGENCE TIME

Classic 802.1D STP takes up to 50 seconds to converge after a topology change (2×15s Listening/Learning + delays). This is unacceptably slow for modern networks — which is exactly why **RSTP (802.1w)** was created.

RSTP — Rapid Spanning Tree

RSTP collapses the state machine to just three states (Discarding, Learning, Forwarding) and introduces edge ports, backup/alternate port roles, and proposal/agreement handshakes — converging in under a few seconds instead of 50.

```
! Enable RSTP and set priority to win root election
Switch(config)# spanning-tree mode rapid-pvst
Switch(config)# spanning-tree vlan 10 priority 4096
Switch(config)# spanning-tree vlan 10 root primary

! Enable PortFast + BPDU Guard on access ports to end devices
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpduguard enable
```

```
Switch# show spanning-tree vlan 10
VLAN0010
  Root ID    Priority    4096
             Address     aaaa.bbbb.cccc
  Bridge ID  Priority    4096 (priority 4096 sys-id-ext 10)
Interface    Role      Sts Cost   Prio Type
-----
Gi0/1        Root     FWD  4      128   P2p
Gi0/2        Desg     FWD  4      128   P2p
```

EXAM TIP

PortFast is only for ports connecting to end hosts (PCs, printers) — it skips Listening/Learning and goes straight to Forwarding. Never enable PortFast on a port connecting to another switch, or you risk a temporary loop. Pair it with **BPDU Guard**, which err-disables the port if it ever receives a BPDU (a sign a switch, not a host, got plugged in).

LAB 9.1 — STP ROOT BRIDGE CONTROL

1. Build a triangle of three switches with redundant links.
2. Observe the default root election with `show spanning-tree`.
3. Force SW1 to become root using `spanning-tree vlan 1 priority 0`.
4. Shut down the root port on a non-root switch and observe convergence time and the new active path.



10 EtherChannel (LACP/PAgP)

Instead of letting STP block a redundant link entirely, EtherChannel bundles 2–8 physical links into one logical link — full bandwidth, load-balanced, with automatic failover, and STP treats the bundle as a single link (no blocking needed).

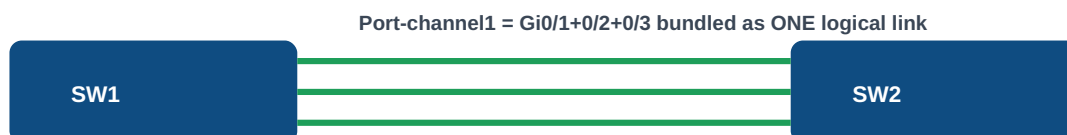


Figure 10.1 — Three physical links bundled into a single EtherChannel, avoiding STP blocking.

Protocol	Vendor	Modes
LACP (802.3ad)	Open standard	active / passive
PAgP	Cisco proprietary	desirable / auto
Static (on)	Any	on — no negotiation, both sides must match manually

```
Switch(config)# interface range gi0/1 - 3
Switch(config-if-range)# channel-group 1 mode active
Switch(config-if-range)# exit
Switch(config)# interface port-channel 1
Switch(config-if)# switchport mode trunk
```

COMMON PITFALL

Every physical port in the bundle must match: same speed, duplex, VLAN/trunk settings, and mode.

Mismatches prevent the bundle from forming (`show etherchannel summary` shows individual ports stuck, not bundled).

11 Switch Security Essentials

Port Security

Restrict which (and how many) MAC addresses can connect on an access port — a first line of defense against rogue devices and MAC flooding attacks.

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security mac-address sticky
Switch(config-if)# switchport port-security violation restrict
```

Violation Mode	Behavior
Protect	Drops offending traffic silently, no log
Restrict	Drops offending traffic, logs and increments counter
Shutdown (default)	Err-disables the entire port — requires manual/automatic recovery

Other essential hardening

- **DHCP Snooping** — trusts DHCP replies only from designated uplink ports, blocking rogue DHCP servers.
- **Dynamic ARP Inspection (DAI)** — validates ARP packets against the DHCP snooping binding table to prevent ARP spoofing.
- **Disable unused ports** and place them in an unused/black-hole VLAN.
- **Disable CDP/LLDP** on ports facing untrusted networks to avoid information leakage.

LAB 11.1 — PORT SECURITY

1. Enable port security on Fa0/1 with a max of 1 MAC, sticky learning, violation mode `shutdown` .
2. Connect PC-A, verify the MAC is learned with `show port-security interface fa0/1` .
3. Connect a second device to the same port and confirm the port goes into `err-disabled` state.
4. Recover the port with `shutdown / no shutdown` .

Routing Fundamentals



12 Router Fundamentals & the Routing Table

A router's job: receive a packet, examine the destination IP, consult the routing table, and forward out the correct interface toward the next hop. Every entry in a routing table answers one question — "to reach this network, go this way."

```
Router# show ip route
Codes: C - connected, S - static, O - OSPF, D - EIGRP, R - RIP

C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
S    10.0.0.0/8 [1/0] via 192.168.10.254
O    172.16.0.0/16 [110/20] via 192.168.10.2, 00:04:12, Gi0/0
D    10.20.0.0/16 [90/3072] via 192.168.10.3, 00:01:05, Gi0/1
```

Reading a route entry: `O 172.16.0.0/16 [110/20] via 192.168.10.2` means: learned via OSPF, administrative distance 110, metric 20, reachable via next-hop 192.168.10.2.

Route selection logic

1. **Longest prefix match wins** — a /32 route beats a /24 route beats a /0 (default route) for the same destination, regardless of protocol.
2. Among routes to the same prefix, the route with the lowest **Administrative Distance (AD)** wins (see Chapter 17 table).
3. Among routes from the same protocol, the lowest **metric** wins.

EXAM TIP

Longest prefix match ALWAYS takes priority over administrative distance. A static /32 route and an OSPF /24 route to overlapping space — the /32 wins even though OSPF might otherwise appear "more trusted."

13 Static Routing

Static routes are manually configured, predictable, and use zero CPU/bandwidth for calculation — ideal for stub networks, small topologies, or backup paths.

```
! Standard static route
Router(config)# ip route 172.16.0.0 255.255.0.0 192.168.10.2

! Default route — "gateway of last resort"
Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1

! Floating static route — backup path with higher AD (200) than a dynamic route
Router(config)# ip route 172.16.0.0 255.255.0.0 192.168.20.2 200
```

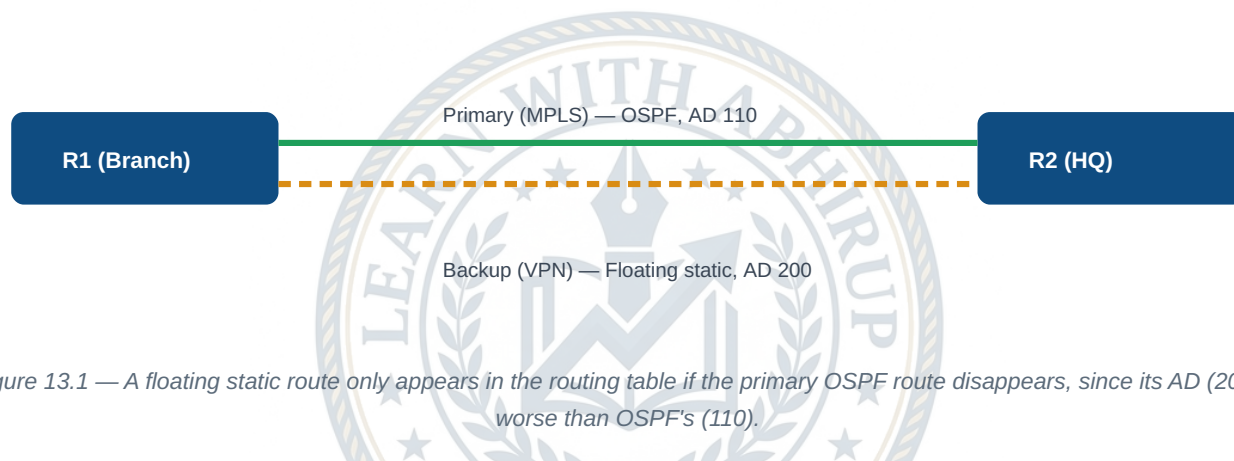


Figure 13.1 — A floating static route only appears in the routing table if the primary OSPF route disappears, since its AD (200) is worse than OSPF's (110).

LAB 13.1 — STATIC + FLOATING STATIC

1. Configure a primary route via OSPF or a directly reachable next hop.
2. Add a floating static route to the same destination with AD 200 via a backup path.
3. Confirm with `show ip route` that only the primary appears.
4. Shut down the primary path's interface and confirm the floating static now appears and traffic reroutes.

14 Dynamic Routing Concepts

Dynamic routing protocols automatically discover networks and adapt to topology changes. They fall into two broad families:

Type	Logic	Examples
Distance Vector	"Routing by rumor" — trusts neighbor's advertised distance	RIP, EIGRP (hybrid)
Link State	Builds a full topology map, runs SPF (Dijkstra) locally	OSPF, IS-IS

RIP — quick overview (legacy, still testable)

RIP is the oldest IGP still occasionally seen — hop-count metric (max 15 hops), slow convergence, simple configuration. Rarely used in production today, but useful for grasping distance-vector basics before OSPF.

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.10.0
Router(config-router)# no auto-summary
```



15 OSPF (Single & Multi-Area)

OSPF (Open Shortest Path First) is the most heavily tested dynamic routing protocol on the CCNA — an open-standard link-state protocol that builds a complete map of the network topology and calculates the shortest path using Dijkstra's algorithm.

OSPF neighbor states

State	Meaning
Down	No hellos received yet
Init	Hello received, but router's own ID not seen in it yet
2-Way	Bidirectional communication confirmed — DR/BDR election happens here
ExStart / Exchange	Negotiating master/slave, exchanging DBD packets
Loading	Requesting missing LSAs
Full	Fully synchronized link-state databases — normal steady state

Basic single-area configuration

```
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router(config-router)# passive-interface gigabitethernet0/1
```

NOTE — WILDCARD MASKS

OSPF's **network** command uses a **wildcard mask**, the inverse of a subnet mask. For a /24, the wildcard is 0.0.0.255. Wildcard = 255.255.255.255 minus the subnet mask.

The OSPF cost metric

OSPF cost = Reference Bandwidth ÷ Interface Bandwidth (default reference bandwidth is 100 Mbps). This means Gigabit and 10-Gigabit links can end up with the same cost (1) unless you raise the reference bandwidth — a well-known gotcha on modern networks.

```
! sets reference to 10 Gbps (value is in Mbps)
Router(config-router)# auto-cost reference-bandwidth 10000
```

Multi-area OSPF

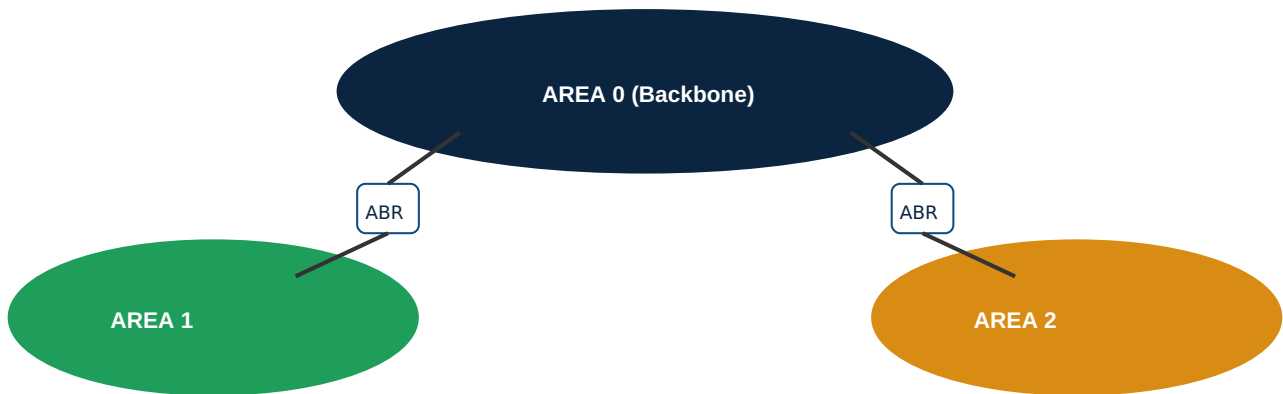


Figure 15.1 — Multi-area OSPF: every non-backbone area must connect to Area 0 through an ABR (Area Border Router), reducing LSA flooding and improving scalability.

Router Role	Description
Internal Router	All interfaces in one area
ABR (Area Border Router)	Interfaces in Area 0 and at least one other area
ASBR (AS Boundary Router)	Redistributes routes from outside OSPF (e.g., static, BGP)
Backbone Router	At least one interface in Area 0

DR/BDR election on multi-access segments

On Ethernet segments, OSPF elects a Designated Router (DR) and Backup DR (BDR) to reduce the number of adjacencies (n routers would otherwise need $n \times (n-1)/2$ adjacencies). Election is based on highest OSPF priority, then highest Router ID.

```
! guarantees this router wins the DR election
Router(config-if)# ip ospf priority 255
```

Verification commands

```
Router# show ip ospf neighbor
Router# show ip ospf interface brief
Router# show ip protocols
Router# show ip route ospf
```

EXAM TIP

For two routers to form an OSPF neighbor relationship, these MUST match: Hello/Dead timers, Area ID, Area type (stub flags), subnet mask (on the segment), and MTU. Authentication type/key must also match if used.

LAB 15.1 — SINGLE-AREA OSPF

1. Configure three routers in a triangle, all in Area 0.
2. Advertise all connected networks with correct wildcard masks.
3. Verify full adjacency with `show ip ospf neighbor` (state = FULL).
4. Confirm end-to-end reachability and inspect the OSPF-learned routes with `show ip route ospf`.



EIGRP (Enhanced Interior Gateway Routing Protocol) is Cisco's advanced distance-vector (technically "advanced distance vector" / hybrid) protocol. It converges extremely fast using the DUAL algorithm and pre-computed backup routes (feasible successors).

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.10.0 0.0.0.255
Router(config-router)# no auto-summary
Router(config-router)# eigrp router-id 1.1.1.1
```

Key EIGRP terms

Term	Meaning
Successor	The best (lowest-metric) route to a destination — installed in the routing table
Feasible Successor	A pre-computed loop-free backup route, instantly promoted if the successor fails
Reported Distance (RD)	A neighbor's own metric to reach a destination
Feasibility Condition	A backup qualifies as feasible successor only if its RD < the successor's total metric

EIGRP's composite metric factors in bandwidth and delay by default (load and reliability are configurable but rarely used). Its fast convergence via pre-computed feasible successors is its headline advantage over OSPF's full SPF recalculation.

```
Router# show ip eigrp neighbors
Router# show ip eigrp topology
Router# show ip route eigrp
```

EXAM TIP

EIGRP is Cisco-proprietary (though a limited open version exists) — a key differentiator from open-standard OSPF. Both are commonly compared on the exam: OSPF = link-state, area-based; EIGRP = advanced distance-vector, autonomous-system based.

17 Route Redistribution & Administrative Distance

Real networks often run more than one routing protocol (e.g., during a migration, or joining two companies). Redistribution injects routes learned by one protocol into another.

Source	Default AD
Directly connected	0
Static route	1
EIGRP (internal)	90
OSPF	110
RIP	120
EIGRP (external)	170
Unreachable/unknown	255 (never used)

EXAM TIP — LOWER AD WINS

Administrative Distance is only compared when two DIFFERENT protocols advertise the SAME prefix length to the SAME destination. Lower AD = more trusted = installed in the routing table.

```
! Redistribute OSPF routes into EIGRP, and vice versa, on a boundary router
Router(config)# router eigrp 100
Router(config-router)# redistribute ospf 1 metric 10000 100 255 1 1500
Router(config)# router ospf 1
Router(config-router)# redistribute eigrp 100 subnets
```

ROUTING LOOP RISK

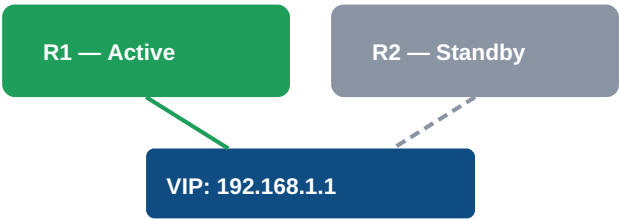
Two-way redistribution on more than one boundary router can create routing loops and suboptimal paths if not carefully filtered with route maps and distribute lists. Always redistribute with a clear plan, not by trial and error.

Redundancy, ACLs, NAT & DHCP



18 First-Hop Redundancy — HSRP & VRRP

End hosts configure exactly one default gateway. If that gateway router fails, hosts lose connectivity — unless a First-Hop Redundancy Protocol (FHRP) is running, presenting a shared **virtual IP** that fails over transparently between physical routers.



Hosts point default gateway to 192.168.1.1 — never a physical router IP

Figure 18.1 — HSRP: hosts use a shared virtual IP; if the active router fails, the standby takes over transparently.

Protocol	Vendor	Roles	Default Virtual MAC
HSRP	Cisco proprietary	Active / Standby	0000.0C07.ACxx
VRRP	Open standard	Master / Backup	0000.5E00.01xx
GLBP	Cisco proprietary	AVG / AVF — allows load balancing	0007.B4xx.xxxx

```
! HSRP configuration on R1 (Active)
R1(config)# interface gi0/0
R1(config-if)# ip address 192.168.1.2 255.255.255.0
R1(config-if)# standby 1 ip 192.168.1.1
R1(config-if)# standby 1 priority 110
R1(config-if)# standby 1 preempt
```

EXAM TIP

`standby preempt` is required if you want a higher-priority router to reclaim the Active role automatically once it comes back online — without it, the current Active stays Active even after the preferred router returns.

19 Access Control Lists (ACLs)

ACLs filter traffic based on rules evaluated top-down, first match wins, with an implicit `deny all` at the end of every ACL.

Type	Number Range	Filters On	Placement Best Practice
Standard	1–99, 1300–1999	Source IP only	Close to the destination
Extended	100–199, 2000–2699	Source/dest IP, protocol, port	Close to the source
Named	N/A (name-based)	Standard or extended rules	Same rules, easier to edit

```
! Named extended ACL – block Telnet from a subnet, allow everything else
Router(config)# ip access-list extended BLOCK-TELNET
Router(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 any eq 23
Router(config-ext-nacl)# permit ip any any
Router(config)# interface gi0/0
Router(config-if)# ip access-group BLOCK-TELNET in
```

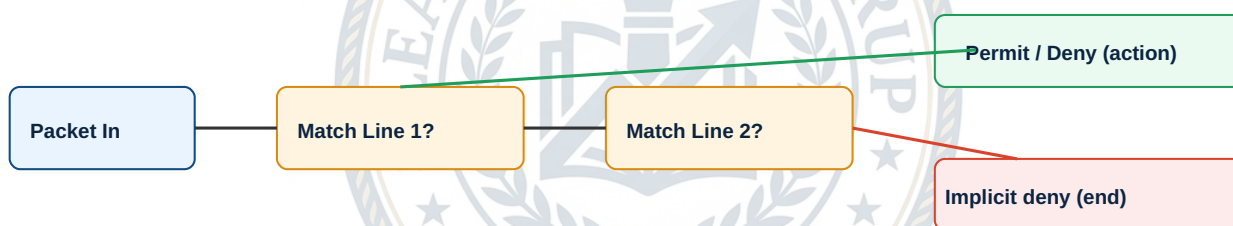


Figure 19.1 — ACL processing: top-to-bottom, first match wins; anything not explicitly matched hits the implicit deny.

COMMON PITFALLS

- 1) Standard ACLs should be placed close to the *destination* since they can only match source IP.
- 2) Every ACL needs at least one `permit` statement, or it silently denies all traffic.
- 3) Order matters — a broad `deny` placed above a specific `permit` will shadow it.

Network Address Translation (NAT)

NAT translates private (RFC 1918) addresses into public, routable addresses — conserving IPv4 address space and adding a layer of obscurity for internal hosts.

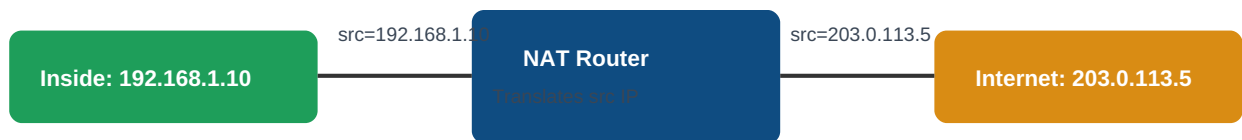


Figure 20.1 — PAT (NAT overload): the router rewrites the source IP (and port) as traffic exits to the internet.

Type	Mapping	Use Case
Static NAT	1-to-1, fixed	Publicly-facing servers needing a consistent public IP
Dynamic NAT	Pool of public IPs, first-come-first-served	Rarely used today — limited by pool size
PAT / NAT Overload	Many-to-1 using port numbers	Standard for internet-edge routers — most common in practice

```
! PAT (NAT overload) configuration
Router(config)# ip access-list standard NAT-SOURCE
Router(config-std-nacl)# permit 192.168.1.0 0.0.0.255
Router(config)# interface gi0/0
Router(config-if)# ip nat inside
Router(config)# interface gi0/1
Router(config-if)# ip nat outside
Router(config)# ip nat inside source list NAT-SOURCE interface gi0/1 overload
```

```
Router# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp  203.0.113.5:1024    192.168.1.10:1024 198.51.100.5:80    198.51.100.5:80
```

EXAM TIP

Know the four NAT terms: **Inside local** (private address of inside host), **Inside global** (translated public address), **Outside local** and **Outside global** (usually identical unless outside NAT is also used).

21 DHCP on Routers

Cisco routers can act as full DHCP servers for connected subnets, or relay DHCP requests to a central server across a routed network.

```
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)# ip dhcp pool SALES-POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# lease 7
```

DHCP Relay (ip helper-address)

If the DHCP server lives on a different subnet, the local gateway router forwards (relays) broadcast DHCP requests as unicast to the real server.

```
Router(config)# interface vlan 20
Router(config-if)# ip helper-address 10.0.0.50
```

THE DORA PROCESS

DHCP client-server exchange: **D**iscover (broadcast) → **O**ffer (server) → **R**quest (client) → **A**cknowledge (server). All four steps use broadcast until the client has a valid address.

Troubleshooting & Real-World Labs



Symptom	Likely Cause	Check
Two hosts in same VLAN can't ping	Wrong VLAN assignment, cable/port down, port security violation	<code>show vlan brief</code> , <code>show interfaces status</code> , <code>show port-security</code>
Intermittent slowness on one link	Duplex mismatch	<code>show interfaces</code> — look for late collisions, CRC errors
Hosts across switches in same VLAN unreachable	Trunk not passing that VLAN, native VLAN mismatch	<code>show interfaces trunk</code>
Network-wide slowdown / broadcast storm	STP loop, missing redundant-link protection	<code>show spanning-tree</code> , check for err-disabled ports
Port stuck in err-disabled	Port security violation, BPDU Guard trip, or loopback detected	<code>show interfaces status err-disabled</code> , then <code>shutdown / no shutdown</code> to recover

SYSTEMATIC APPROACH

Always work bottom-up through the OSI model for connectivity issues: Physical (link light, cable) → Data Link (VLAN, trunk, STP) → Network (IP addressing, routing) → above. Skipping straight to advanced protocols before confirming Layer 1/2 wastes time.

23 Routing Troubleshooting Playbook

Symptom	Likely Cause	Check
Route missing from routing table	Wrong network statement/wildcard, interface down, mismatched area	<code>show ip route</code> , <code>show ip protocols</code> , <code>show ip ospf interface brief</code>
OSPF neighbors stuck in 2-Way or never form	Area mismatch, hello/dead timer mismatch, ACL blocking multicast	<code>show ip ospf neighbor</code> , <code>debug ip ospf adj</code>
Asymmetric routing / one-way ping	Missing return route, ACL blocking return traffic, NAT misconfig	Trace both directions with <code>traceroute</code>
Static route not appearing	Next hop unreachable, AD beaten by another route	<code>show ip route static</code> , verify next-hop reachability
Redistribution creating routing loops	Two-way redistribution without filtering	Trace path with <code>traceroute</code> , review redistribute filters/route-maps

Core diagnostic commands

```
Router# ping
Router# traceroute
Router# show ip interface brief
Router# show cdp neighbors detail
Router# show running-config
Router# debug ip routing ! use sparingly in production
```

24 End-to-End Lab Scenarios

CAPSTONE LAB A — SMALL OFFICE NETWORK

Topology: 1 router (R1), 2 access switches (SW1, SW2), 3 VLANs (Sales-10, Eng-20, Mgmt-99).

Tasks:

1. Create VLANs on both switches; trunk SW1 ↔ SW2 and SW1 ↔ R1.
2. Configure router-on-a-stick sub-interfaces for all 3 VLANs.
3. Enable RSTP, force SW1 as root with priority 4096.
4. Configure port security on all access ports (max 2 MACs, sticky).
5. Add a default static route from R1 toward a simulated ISP.
6. Verify full connectivity: same-VLAN, cross-VLAN, and outbound.

CAPSTONE LAB B — REDUNDANT CAMPUS CORE

Topology: 2 Layer-3 distribution switches (Dist1, Dist2), 2 access switches, OSPF area 0 throughout, HSRP between distribution switches for each VLAN's default gateway.

Tasks:

1. Configure SVIs for VLAN 10/20 on both distribution switches.
2. Run OSPF between Dist1, Dist2, and a simulated edge router.
3. Configure HSRP per VLAN with Dist1 as Active (priority 110, preempt) and Dist2 as Standby.
4. Bundle the Dist1–Dist2 link as an EtherChannel (LACP active/active).
5. Simulate a Dist1 failure (shutdown its uplink) and verify HSRP failover and OSPF reconvergence.

CAPSTONE LAB C — BRANCH-TO-HQ WITH NAT & ACL

Topology: Branch router with inside LAN (192.168.50.0/24), PAT to a simulated ISP, and a floating static backup route via a secondary link.

Tasks:

1. Configure PAT overload on the primary WAN interface.
2. Write an extended ACL permitting only HTTP/HTTPS/DNS outbound, applied inbound on the LAN interface.
3. Configure a primary static default route and a floating static backup with AD 200.
4. Test failover by shutting down the primary WAN interface.
5. Verify NAT translations with `show ip nat translations` during a live ping/traceroute.

CCNA Exam Preparation



The CCNA exam blueprint weights content roughly as follows (percentages are illustrative of the domain emphasis — always check Cisco's current published blueprint for exact figures):

Domain	Approx. Weight	Chapters in This Course
Network Fundamentals	~20%	1, 2, 3
Network Access (Switching)	~20%	4 – 11
IP Connectivity (Routing)	~25%	12 – 17
IP Services (NAT/DHCP/FHRP)	~10%	18 – 21
Security Fundamentals	~15%	11, 19
Automation & Programmability	~10%	Not covered — separate course topic

NOTE

This course deliberately focuses on Switching and Routing — the two heaviest, highest-leverage domains. Automation/programmability (APIs, JSON, controller-based networking) is a distinct topic worth its own dedicated study.

Switching

```
vlan 10 / name Sales
switchport mode access / trunk
switchport access vlan 10
switchport trunk allowed vlan 10,20
spanning-tree mode rapid-pvst
spanning-tree vlan 10 priority 4096
spanning-tree portfast / bpduguard enable
channel-group 1 mode active
switchport port-security / maximum 2 / mac-address sticky
show mac address-table / show vlan brief / show interfaces trunk
show spanning-tree vlan 10 / show etherchannel summary
```

Routing

```
ip route [dest] [mask] [next-hop] [AD]
router ospf 1 / network [ip] [wildcard] area 0
router eigrp 100 / network [ip] [wildcard]
standby 1 ip [vip] / standby 1 priority 110 / standby 1 preempt
ip nat inside / ip nat outside / ip nat inside source list ... overload
ip dhcp pool NAME / network ... / default-router ...
show ip route / show ip ospf neighbor / show ip eigrp neighbors
show ip nat translations / show ip interface brief
```

Verification & troubleshooting

```
ping / traceroute / show cdp neighbors detail
show running-config / show interfaces [name]
debug ip ospf adj (use sparingly)
```

1. Which device makes forwarding decisions based on destination MAC address?
a) Router b) Switch c) Hub d) Firewall
2. What is the usable host range for 172.16.50.77/28?
a) .64–.79 b) .65–.78 c) .65–.79 d) .64–.78
3. Which STP enhancement should be applied to a port connecting to a PC, but never to an inter-switch link?
a) BPDU Filter b) Root Guard c) PortFast d) Loop Guard
4. What administrative distance does OSPF use by default?
a) 90 b) 100 c) 110 d) 120
5. Which command enables routing between VLAN SVIs on a multilayer switch?
a) `ip routing` b) `router ospf 1` c) `no switchport` d) `ip route 0.0.0.0`
6. A floating static route with AD 200 will be used instead of an OSPF route when:
a) Always, since static beats dynamic b) Never c) Only when the OSPF route disappears from the table d) Only during business hours
7. In PAT, which term describes the translated public address of an inside host?
a) Inside local b) Inside global c) Outside local d) Outside global
8. HSRP requires which command for a higher-priority router to reclaim the Active role automatically?
a) `standby priority` b) `standby preempt` c) `standby track` d) `standby version 2`
9. Which switching method forwards a frame after reading only the destination MAC, without full error checking?
a) Store-and-forward b) Fragment-free c) Cut-through d) Hybrid
10. What does the "implicit deny" at the end of every ACL do?
a) Permits all unmatched traffic b) Denies all unmatched traffic c) Logs but permits traffic d) Only applies to extended ACLs

Answer Key: 1-b, 2-b, 3-c, 4-c, 5-a, 6-c, 7-b, 8-b, 9-c, 10-b

Chapter 2 Subnetting Drill Answers:

- 1) 10.1.1.200/26 → Network: 10.1.1.192, Broadcast: 10.1.1.255, Hosts: .193–.254
- 2) 192.168.5.130/27 → Network: 192.168.5.128, Broadcast: 192.168.5.159, Hosts: .129–.158
- 3) $/29 = 6$ usable hosts ($2^3 - 2$); a /24 contains 32 separate /29 subnets ($256 \div 8$).

End of Course — Switching & Routing Mastery. Keep practicing in Packet Tracer or GNS3/EVE-NG until every configuration in this guide is muscle memory. Good luck on the exam.

